

Geography and admixture shape the genome-scale phylogeny of North American *Delphinium*

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Summary

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- The genus *Delphinium* exemplifies the complexity of plant diversification in mountainous regions, where rapid speciation, hybridization, and morphological convergence frequently obscure species boundaries. The North American lineage *Delphinium* sect. *Diedropetala* has rapidly radiated across a range of ecological habitats, from alpine tundra to desert grasslands, forming an iconic component of western wild flower communities.
- Here, we present a genome-scale phylogeny of 34 taxa using ddRAD-seq data from 150 individuals sampled across their geographic ranges.
- Our results reveal strong phylogenetic structure corresponding to biogeographic regions, including several well-supported clades that cut across existing taxonomic subsections, emphasizing the need for taxonomic revision. Hybridization and introgression are widespread, occurring both within and between these regional clades. Most species are connected through a syngameon-like network of introgression, with few widespread species acting as central hubs. We describe six higher-level clade names and use an agent-based artificial intelligence analysis to identify morphological synapomorphies that align with our genomic findings.
- Our study demonstrates that geography and gene flow have played a dominant role in shaping the evolutionary history of *Delphinium* in North America, offering a framework for revising its taxonomy and informing future conservation efforts.

Introduction

Western North America harbors exceptional topographic and ecological heterogeneity that has shaped the diversification of many endemic plant lineages. Its flora includes numerous rare and specialized taxa narrowly distributed along climatic gradients, many of which now face mounting threats from shifting climate and precipitation patterns, increased wildfire frequency, and habitat loss (Overpeck & Udall, 2020; Kannenberg *et al.*, 2021). Although North America is among the best-studied floras globally (Cronquist, 1984; Hickman, 1993; Flora of North America, 1997), persistent taxonomic uncertainties continue to impede efforts to quantify biodiversity and track its decline. These knowledge gaps limit our ability to detect local extinctions (Wiens, 2016), estimate species range shifts (Frei *et al.*, 2010), and prioritize conservation actions.

The delimitation of species and their phylogenetic relationships remains poorly understood even within some of the most enigmatic plant lineages of this region, including larkspurs (*Delphinium*; Jabbour & Renner, 2012; Xiang *et al.*, 2017), beardtongues (*Penstemon*; Wessinger *et al.*, 2016; Wolfe *et al.*, 2021), and paintbrushes (*Castilleja*; Jacobs *et al.*, 2019; Tank *et al.*, 2009). This is especially true within applications of morphological systematics, where lineages may be under- or over-split in relation to their rates of

morphological evolution. In montane regions, these rates can vary considerably. Geographically isolated populations or species are prone to under-counting when they exhibit parallel evolution or cryptic variation (e.g. Donoghue *et al.*, 2022); whereas taxa may be over-split when populations exhibit extensive environmental plasticity, or stable hybrid phenotypes (DiVittorio *et al.*, 2020). Integrative systematic studies that incorporate phylogenomic evidence alongside morphological and geographic information can better assess the distinctiveness of taxa, especially in lineages characterized by hybridization, plasticity, and parallelism (e.g. Hipp *et al.*, 2020; Donoghue *et al.*, 2022). Such work contributes not only to improving plant taxonomy and biodiversity research but also to detecting associations between environments and the evolution or expression of unique phenotypes (Anstett *et al.*, 2021; Melton *et al.*, 2021; Gross *et al.*, 2024).

A prime example of these dynamics is the North American clade *Delphinium* sect. *Diedropetala* Huth. After originating in the Himalaya-Hengduan region of Asia *c.* 32 million years ago (Ma), *Delphinium* L. diversified into 650–700 species, dispersing and speciating west through Eurasia, the Mediterranean, and into Africa, and east through eastern Asia, only recently arriving in western North America *c.* 3 Ma, where it rapidly radiated into *c.* 60 species (Jabbour & Renner, 2012). While *Delphinium* exhibits both annual and perennial life cycles, all species of the North

American sect. *Diedropetala* are perennial herbaceous plants characterized by large showy flowers that are typically blue and purple, and less commonly red, yellow, or white. Many species are widespread and abundant in montane, grassland, and desert habitats, while others exist as narrow endemics that have specialized adaptations to harsh alpine environments (Warnock, 1995). Multiple species are currently endangered in North America. They are economically important as ornamentals and intensely studied for their toxicity to livestock (Gardner *et al.*, 2002; Cook *et al.*, 2009, 2017; Pfister *et al.*, 2014). Their rapid diversification in North America has been attributed to dispersal and isolation in the various mountain ranges of the western Cordillera, especially the Cascades, Sierra Nevada, and Rocky Mountains (Warnock, 1995). Despite multiple taxonomic and phylogenetic research efforts focused on this clade, there is no fully resolved phylogeny of *Delphinium* sect. *Diedropetala*, and little is known about its biogeographic history across western North America.

The genus *Delphinium* is often considered taxonomically difficult due to a limited number of distinguishing morphological characters. Two comprehensive taxonomic treatments for North American *Delphinium* have been published over the last century which recognize between 60 and 80 species, but disagree over their exact number, nomenclature, and distribution. In the first published circumscription of the clade, Ewan (1945) combined extensive fieldwork and study of herbarium specimens to distinguish 79 species grouped into 13 'series' based on seed morphology, leaf architecture and pubescence, corolla color, and other anatomical features. Fifty years later, Warnock's treatment (Warnock, 1995) reduced the number of species from 79 to 61, comprising two sections: *Elatopsis* Huth, represented by a garden cultivar (*Delphinium elatum* L.) introduced from Europe, and *Diedropetala*, which is composed of nine subsections. These subsections are largely delineated by root structure and size, sepal color and length, seed morphology, and fruit position. This treatment is included in the current Flora of North America (FNA; Warnock, 1997), and we will refer to it hereafter as the FNA treatment. Other notable regional treatments of *Delphinium* sect. *Diedropetala* included as follows: Ackerfield (2022), Chambers (2018), Cronquist (1984), Ewan (1942), Holmgren *et al.* (2012), Koontz & Warnock (2012), Lewis & Epling (1954), Rydberg (1899), Weber & Wittmann (2012), Wilde (1931).

Two molecular phylogenetic studies have used Sanger sequencing markers to examine relationships within section *Diedropetala*. A study by Koontz *et al.* (2004) sampled 62 species of North American *Delphinium* but found that few relationships can be resolved using a small number of molecular markers. A deeper-scale phylogenetic analysis by Jabbour & Renner (2012) sampled a third of the North American species, as well as many other species from across its global distribution. Their results support section *Diedropetala* as a monophyletic clade, but similarly lacked support to confidently resolve many relationships within it. While more recent molecular studies exist for *Delphinium* clades in east Africa (Chartier *et al.*, 2016), the Mediterranean (Xiang *et al.*, 2017), Russia (Kashin *et al.*, 2021), and China (Song *et al.*, 2024), no study has yet brought genome-scale data

to bear on phylogenetic relationships within *Delphinium* sect. *Diedropetala* in North America.

Multi-locus phylogenomic analyses offer an advantage not only for inferring species tree relationships from among discordant (conflicting) phylogenetic signals, but additionally for the potential to test for evidence of hybrid introgression (Durand *et al.*, 2011; Eaton *et al.*, 2015; Kubatko & Knowles, 2023). Hybridization is likely to have played a significant role throughout the diversification of *Delphinium*. Many putative hybrids have been described based on morphological descriptions (Ewan, 1945; Legro, 1961; Warnock, 1995; Koontz *et al.*, 2001), but the extent to which they contribute to introgressive gene flow between species remains unknown. Our primary objectives in this study were to use genome-wide data to resolve the phylogenetic backbone of *Delphinium* sect. *Diedropetala* and investigate the frequency of introgressive hybridization with respect to species barriers and geography. To achieve this, we generated a new reduced-representation genomic dataset to investigate phylogeny and introgression across *Delphinium* sect. *Diedropetala*. We present a new and highly revised phylogenomic hypothesis for relationships among species representing all previously described subsections in this group, and discuss our confidence in this result with regard to our extensive sampling of populations within species, tests for hybrid introgression, and the biogeographic concordance of our results.

Materials and Methods

Taxon sampling

We collected voucher specimens and silica-preserved leaf samples during field expeditions between 2019 and 2024. Additional samples were provided by the USDA Poisonous Plant Research Lab as part of a long-term research program concerning toxicity in *Delphinium* L., where tissues were stored freeze-dried. This dataset includes 150 individuals, with replicate samples of 34 *Delphinium* taxa collected from multiple geographically disjunct populations. When possible, we collected species from their type specimen localities, and our sampling includes at least one taxon as a representative of each subsection in the FNA treatment of *Delphinium*, including the type species for subsections *Bicoloria* (Rydberg) N. I. Malyutin, *Exaltata* N. I. Malyutin, *Multiplex* M. J. Warnock, and *Subscaposa* Ewan. We included samples from two *Aconitum* L. species as outgroups. All collections have associated geographic coordinates (Supporting Information Table S1) and voucher specimens available at the USDA Poisonous Plant Research Lab Herbarium (PPRL) in Logan, UT.

DNA sequencing and bioinformatics

Genomic DNA was extracted from lyophilized leaf tissue using the Quick DNA Extraction Kit (Zymo Research, Irvine, CA, USA) adapted to a 96-well format. DNA quantity and quality were assessed using a spectrophotometer, TapeStation (Agilent), and gel electrophoresis, and genomic DNA was standardized to 20 ng μL^{-1} . Genomic double-digest restriction-site associated

DNA (ddRAD-seq; Peterson *et al.*, 2012) libraries were prepared at the USDA Forage & Range Research Lab at Utah State University using the PstI and MspI restriction enzymes (Poland *et al.*, 2012), followed by adapter ligation, sample pooling, and a final purification and partial polymerase chain reaction conducted for 16 cycles. The pooled library was sequenced on an Illumina Nextseq 2000 at the Utah State University Center for Integrated Biosystems (Logan, UT, USA) to produce single-end 75-bp reads. Sequence data for all samples were assembled in *ipyrad* v.0.9.104 (Eaton & Overcast, 2020) using the *de novo* pipeline and default parameter settings, except for the following changes: `datatype = ddRAD`, `restriction_overhang = ("TGCAG")`, `filter_adapters = 2`, `max_depth = 10 000`, `max_SNPs_locus = 0.4`, `max_shared_Hs_locus = 6`, and `max_alleles_consens = 4`. Additional downstream analyses used the *ipyrad-analysis* toolkit to subsample and convert data to the appropriate format for various analysis tools while also examining the impact of missing data.

Phylogenetic inference

We inferred phylogenetic relationships at both the individual and population-level using multiple inference methods, including concatenation, SNP-based, and gene-tree based. We generated two concatenated supermatrix datasets comprising either all loci shared across at least 4 samples ('min4' dataset) or 10 samples ('min10' dataset), using the *ipyrad-analysis* `window_extractor` tool (Eaton & Overcast, 2020). For each dataset, we inferred a maximum likelihood phylogenetic tree in *raxml-ng* v.1.2.2 (Kozlov *et al.*, 2019) using the GTR + G substitution model with four rate categories, starting from 10 random and 10 parsimony starting trees, and with 200 nonparametric bootstrap replicates. Because concatenation can lead to biased inference in the presence of incomplete lineage sorting, we also employed coalescent-based species tree inference approaches. First, we inferred a species tree from each dataset using *caster* v.1.19 (Zhang *et al.*, 2025b), a single nucleotide polymorphism (SNP)-based approach for inferring species trees from quartet relationships. We used the 'caster-site' method which scores topologies for each site pattern shared among four taxa and infers a phylogeny by combining the quartet trees with highest support. We set the chunk size parameter to 500 and designated for it to incorporate heterozygous variant calls. Support was assessed from 1000 local block bootstraps.

To achieve a 'population-level' phylogeny, we additionally ran *caster* using the same settings as above but with a mapping file to assign individuals to populations. We included 41 populations, where each represents a unique clade corresponding to a taxon name that was consistently recovered as monophyletic in all phylogenetic analyses. Two clades did not meet this criterion. We assigned the outgroup sample Acol2-3 to *A. delphiniifolium* D.C. even though it consistently groups with low support between the two outgroup clades. We assigned the sample Dalp4-2 to *D. ramosum* Rydberg population 4–10, since it consistently groups within that clade, despite being initially determined as *D. alpestre* Rydberg.

We also inferred individual gene trees for RAD-seq loci to use as input to the species tree program ASTRAL IV v.1.23.4.6

(Zhang *et al.*, 2025a). For this analysis, we selected a subset of individuals that had the greatest number of assembled loci from each of the 41 populations identified above. This allowed us to select a subset of loci that were shared across these samples and contained sufficient information for gene tree inference. For these samples, we used the *ipyrad-analysis* `tree_slider` tool to select all loci that contained at least 10 samples from this subset, and a minimum of five variant sites after pruning all other samples. For each locus, we inferred an ML tree in *raxml-ng* as described above. We collapsed all edges with bootstrap support <50%, and input the set of trees to ASTRAL to infer a species tree under default settings. The Python package TOY TREE (Eaton, 2020) was used to collapse low support edges, compute tree distance metrics, and visualize trees.

Introgression analyses

We calculated D-statistics (Durand *et al.*, 2011) and F4 ratios using the Dtrios method in Dsuite (Malinsky *et al.*, 2021), and plotted admixture proportions on species tree hypotheses using the f-branch statistic method. This requires a species tree hypothesis, and so we performed the test on three trees, using the population-level tree inferred by *caster*, as well as population-level trees matching the ML 'min4' and 'min10' topologies, created by similarly collapsing monophyletic clades into a single tip. Analyses were run using a VCF file from the 'min10' assembly, and with the parameter `-k 500` to set the number of jack-knife blocks. For all tests, we used the pooled allele frequencies among the outgroup samples of *Aconitum* as the P4 taxon. We also used allele frequencies among pooled replicate samples from monophyletic species, or populations, to represent the P1, P2, or P3 taxa.

Investigating hybrid complexes

To investigate nonmonophyletic taxon relationships in a complex formed by *D. robustum* Rydberg, *D. ramosum*, and *D. alpestre*, we performed additional phylogenetic and principal component analyses (PCAs). We used the *ipyrad-analysis* `window_extractor` tool to extract a sequence alignment that includes data only for taxa in this subclade (18 samples), in addition to two outgroup taxa. Here, we used *D. californicum* Torrey & A. Gray and *D. hesperium* A. Gray as more closely related outgroups than *Aconitum*, since these taxa showed no evidence of admixture with other taxa, and are suitable to serve as outgroups for this analysis. These alignments include all sites with data shared across at least 10 samples (min10) and were input to *raxml-ng* with the same settings described above to infer ML trees. To examine population structure, we used the *ipyrad-analysis* `pca` tool to extract unlinked SNPs and perform PCA. We conducted this analysis on both the focused dataset (18 samples) and on a larger dataset that includes other closely related species and potentially admixed species (39 samples). For each analysis, we retained only bi-allelic SNPs with zero missing data across the subset of samples examined and randomly subsampled one SNP per RAD locus to reduce the effect of linkage. To examine variation over the randomly sampled subset of unlinked SNPs, we performed 50

replicate analyses and plotted the centroid point for each sample over a low opacity cloud of points from all replicates. Results were contextualized relative to the geographic distribution of samples examined in interactive Folium maps generated with the Esri.WorldTopoMap base layer (Sources: Esri, HERE, Garmin, Intermap, INCREMENT P, GEBCO, USGS, FAO, NPS, NRCAN, GeoBase, IGN, Kadaster NL, Ordnance Survey, Esri Japan, METI, Esri China (Hong Kong), © OpenStreetMap contributors, GIS User Community).

Taxonomy

We propose a set of formal taxonomic names, following the Phylocode protocol (de Queiroz *et al.*, 2020), for the six higher-level clades containing more than one taxon that were consistently recovered across our phylogenomic analyses (Notes S1). To identify additional supporting evidence for these clades beyond genomics, we employed OpenAI's ChatGPT (GPT-4) in a tool-enabled agent-based framework to retrieve and analyze morphological descriptions for each taxon. The agent performed structured searches to retrieve data from the FNA (Wernick, 1997) and other authoritative referenced sources, fetching details on stem height, leaf morphology, inflorescence size, sepal color, spur length, and habitat. After collating these character states, it identified synapomorphies that unite members of our specified clades, and drafted clade definitions that highlight diagnostic characters. All data-based statements are linked to references. The process was iterative with feedback from the authors to generate etymologically appropriate names, revise formatting, and suggest additional comparisons. Manual measurements of herbarium specimens were conducted to verify that morphological characters extracted by the agent were consistent for the newly defined clades.

Results

Genomic ddRAD sequencing and assembly

Genomic sequencing yielded a mean of 1071 118 ($\pm$ SD 712 371) raw reads across all samples, which were assembled into relatively similar numbers of *de novo* clusters within each sample (mean $\pm$ SD = 19 261 $\pm$ 7331). We dropped samples with fewer than 1000 clusters. After clustering across the remaining samples and applying filters, the 'min4' dataset included 80 169 loci, with the total alignment size of 150 samples $\times$ 6490 732 sites (85.93% missing) and contained 556 493 SNPs (68.80% missing). The 'min10' dataset included 29 196 loci with a total alignment size of 150 samples $\times$ 2652 279 sites (70.96% missing) and contained 399 001 SNPs (58.03% missing).

Phylogenetic inference recovers monophyletic taxa

All phylogenetic analyses recovered topologies showing individuals of the same species consistently grouped into monophyletic clades, with a few exceptions observed within *D. distichum* Geyer ex A. Gray, *D. nuttallianum* Pritzl, *D. ramosum*, and *D. alpestre*

(addressed below in the context of admixture). *Caster* recovered the same tree topology from both the 'min4' and 'min10' datasets (Figs 1, S1), while the maximum likelihood trees inferred by *raxml-ng* differed from each other, and from the *caster* topology. These differences are concentrated across the backbone of the tree, in the order of relationships among several major clades, which generally received low support across all trees (Figs S2, S3). Tree distances, measured using the generalized Robinson–Foulds mutual clustering information metric (MCI; Smith, 2020), show that the two ML topologies share 93.3% of information in common, while the *caster* topology shares 86.0 and 91.3% of information in common with the ML 'min4' and 'min10' trees, respectively.

Among our two analyses that grouped individuals into 41 populations, or subsampled a single individual from each of those populations, the resulting topologies were similar. The SNP-based *caster* population analysis recovered the same relationships as in the *caster* analysis of all individuals. The ASTRAL species tree analysis based on inferred gene trees also recovered a similar topology, but with many internal edges receiving < 50% support (Fig. S4). This analysis was based on 3121 gene trees that passed our filtering thresholds, but the information content in individual gene trees was very low. The average number of internal edges per tree was only 2.18 after collapsing edges with < 50% support. Therefore, we put greater confidence in the *raxml-ng* and *caster* results.

All of our tree hypotheses support three early diverging lineages that split from the rest of our sampled *Delphinium* taxa early in the radiation. This includes *D. patens* Benth, *D. nudicaule* Torr. & A. Gray, and *D. trolliifolium* A. Gray, which are distributed in central California, northern California, and northern California, and the Pacific Northwest, respectively. We focus most of our further analyses and discussion on the larger recovered clades in our study.

Higher-level clades correspond strongly with biogeographic regions

A set of six higher-level clades was recovered across all trees that show a strong concordance with the following biogeographic regions: GB1 (Great Basin), PNW (Pacific Northwest), CAL (California), IMW (Intermountain West), ALCAN (Alaska/Canada), and SW (Southwestern United States; Fig. 2a,b). None of these clades corresponds uniquely with an existing taxonomic subsection in the FNA treatment (Fig. 2b). Support for monophyly of each of these clades is > 95% in all trees, despite the fact that their ordering varies and has low support among trees (Fig. 2c–f). In addition to these six monophyletic clades, we also designated two other biogeographic regions, namely GB2 (Great Basin) and ENA (eastern North America), but do not consider these as monophyletic clades because they are each represented by a single species only. The grouping of regions IMW, ALCAN, ENA, and SW is consistently recovered with 100% support, uniting a clade that extends from Alaska to the southern Rocky Mountains. By contrast, the phylogenetic placement of clades CAL, GB1, and GB2 has lower support and is more variable among our inferred trees.

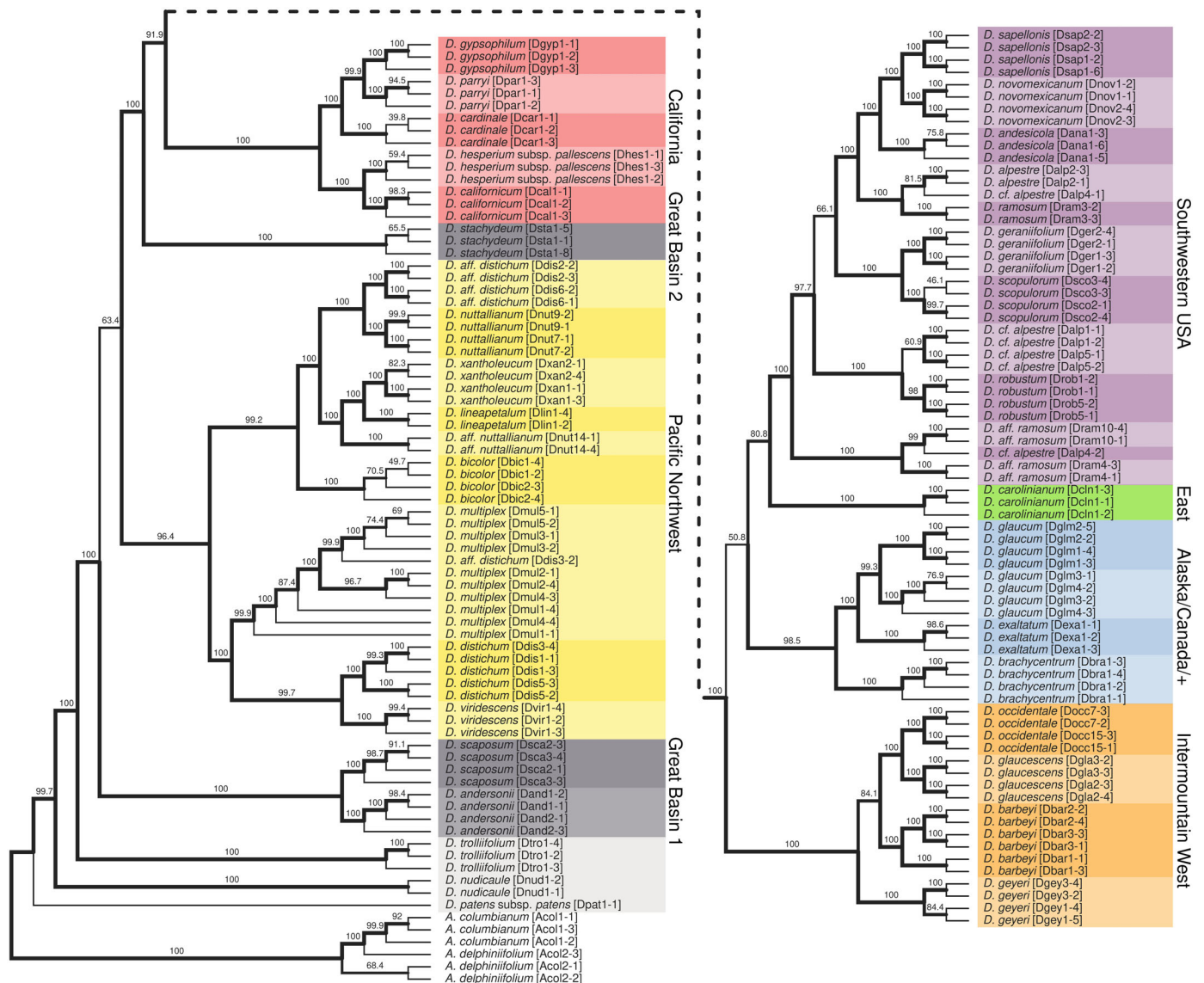


Fig. 1 Rooted phylogenetic tree for 150 samples of *Delphinium* sect. *Diedropetala* inferred from the 'min10' single-nucleotide polymorphism (SNP) dataset using *caster*. Colored rectangles highlight groups that correspond well with distinct geographic regions, representing higher-level clades. Bracketed names to the right of tips refer to sample accession numbers (see Supporting Information Table S1). Populations and individuals sampled from within the same taxon designation form monophyletic clades, except in a few cases involving *Delphinium distichum*, *Delphinium nuttallianum*, *Delphinium ramosum*, and *Delphinium alpestre*.

The GB1 clade occurs in the Great Basin and is represented in our tree by *D. scaposum* Greene sampled from SE Nevada and NW Arizona, and *D. andersonii* A. Gray sampled from southern Idaho. This clade does not group with the only other taxon we sampled from the Great Basin, *D. stachydeum* (A. Gray) Tidestrom, from northern Nevada, which is the sole representative of the biogeographic region designated GB2. This split aligns with morphological differences between these two groups, with *D. stachydeum* exhibiting tall stems (70–150 cm), dense racemes (30–60 flowers), and distinctive fruits when compared to the more diminutive *D. scaposum* (25–50 cm tall with 10–25 flowers per raceme) and *D. andersonii* (30–60 cm tall with 10–25 flowers per raceme). Neither grouping corresponds to an existing

taxonomic subsection, and the subsection names the taxa are currently assigned to are each nonmonophyletic on our phylogeny.

The PNW clade corresponds to taxa distributed in mountainous regions in and around the Columbia Plateau in E Washington, NE Oregon, ID, and W Montana. It includes *D. lineapetalum* Ewan, *D. xantholeucum* Piper, *D. nuttallianum*, *D. distichum*, *D. bicolor* Nuttall, *D. viridescens* Leiberg, and *D. multiplex* (Ewan) C. L. Hitchcock. We recover both *D. nuttallianum* and *D. distichum* as polyphyletic, possibly as a result of admixture, which we investigate below. Morphologically, this clade contains several species of short stature (10–60 cm) that have basal leaves and small to moderate inflorescences (3–50 flowers), but which vary in sepal color, spur length (8–23 mm),

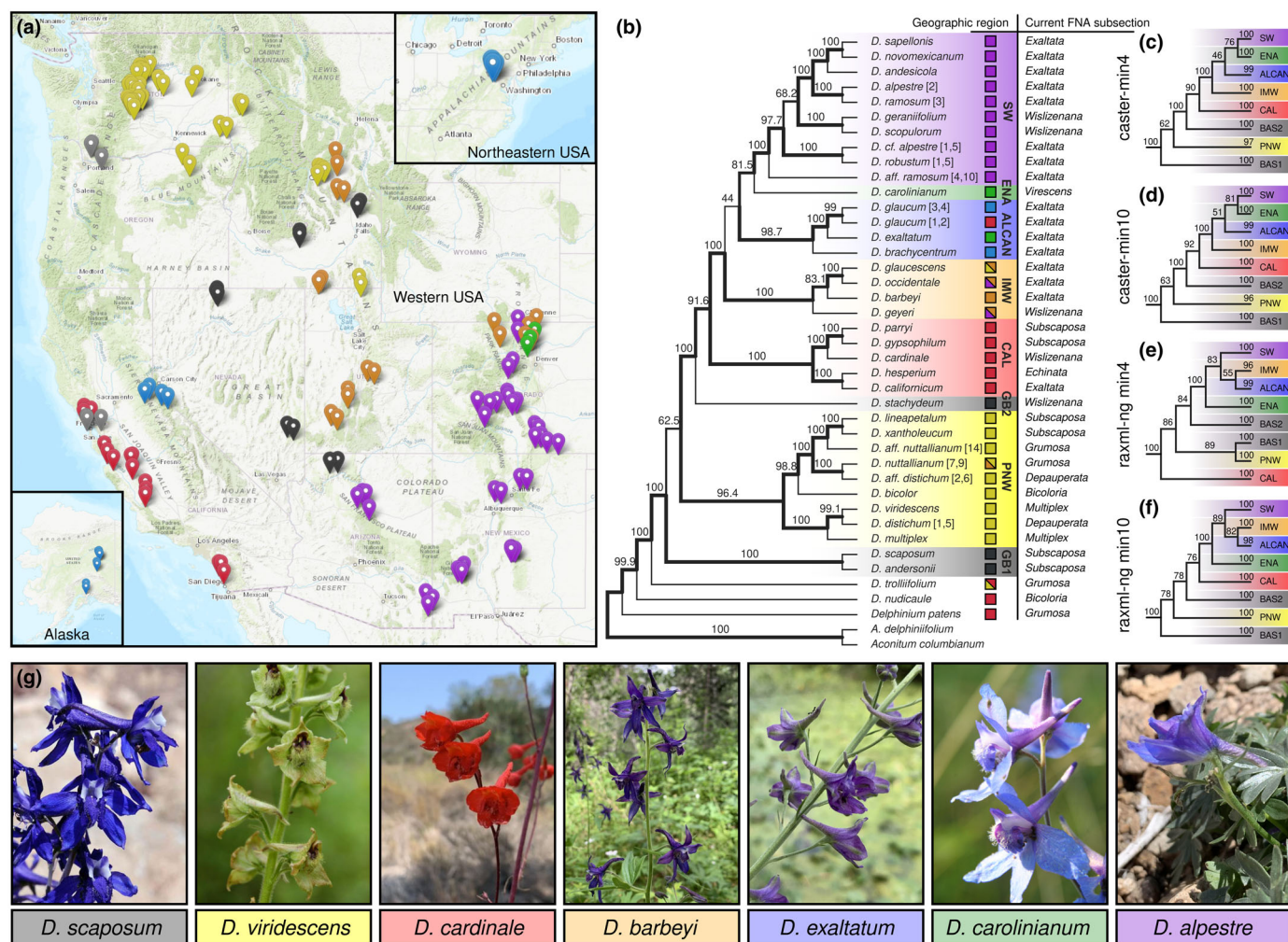


Fig. 2 Phylogeny and biogeographic distribution of *Delphinium* sect. *Diedropetala*. (a) Locations of samples used in this study. Color codes match to those used to represent distinct clades in (b) that correspond closely with the following biogeographic regions (black, Great Basin; yellow, Pacific Northwest; red, California; orange, Intermountain West; blue, Alaska and Canada; green, Eastern United States; purple, Southwestern United States; gray, other). (b) A rooted species tree topology inferred from the 'min10' single-nucleotide polymorphism (SNP) dataset using *caster* with population mapping of 150 individuals to 41 populations, shown with local block bootstrap supports. Nonmonophyletic taxa are indicated by a species-specific population number to the right of the tip name that corresponds to sample accession numbers (Supporting Information Table S1). One or more biogeographic states occupied by each taxon are shown in a colored marker next to each tip. The current subsection designation of each taxon from the Flora of North America is indicated next to each tip. (c–f) There is substantial conflict across the backbone of the tree, which resolves into different topologies when analyzed using different datasets or methods. Support for monophyly of each clade is consistently > 95% (values at the tips of each tree), but support for internal splits between major clades is comparatively low. (g) Images of members of *Delphinium* sect. *Diedropetala*. Colored blocks around species names correspond with clades or biogeographic regions shown in (b), but the species shown are not necessarily representative of the morphological characters of their entire clade or region (photographs of *Delphinium scaposum* and *Delphinium carolinianum* from Lady Bird Johnson Wildflower Center at the University of Texas; *Delphinium viridescens* from the Burke Herbarium at the University of Washington; all others by authors).

and seed morphology. The seven taxa in this clade are grouped into five different subsections in the current FNA treatment, suggesting that our results represent a major departure from their expected relationships based on morphology.

The CAL clade includes taxa distributed primarily in California, represented in our tree by *D. californicum* Torrey & A. Gray, *D. hesperium*, *D. cardinale* Hook., *D. gypsophilum* Ewan, and *D. parryi* A. Gray. These five taxa, which have 100% bootstrap support both as a group, and for each within-species clade, are currently placed in four different subsections in the current FNA treatment. While these species all typically exhibit unwinged

seeds, they have a wide range of stem height (40–160 cm), spur length (7–24 mm), and floral color. The most uniting feature of this clade is their biogeographic distribution centered in California.

The IMW clade corresponds to taxa distributed in the Intermountain West region between the Cascades/Sierra Nevada and the Rocky Mountains. It is represented by four species in our study, *D. geyeri* Greene, *D. barbeyi* (Huth) Huth, *D. occidentale* (S. Watson) S. Watson, and *D. glaucescens* Rydberg. These species tend to have large geographic ranges, occurring in subalpine environments both in this region and surrounding regions. They

all exhibit moderate-to-tall stems (35–150 cm) and lobed leaves (2–9 cm × 3–15 cm), with racemes composed of 6–50 flowers that are purple, blue, or pale-blue with medium-length spurs (9–18 mm).

The ALCAN clade includes *D. brachycentrum* Ledebour, which is endemic to tundra environments in Alaska and northern Asia, and the widespread species *D. glaucum* S. Watson, a tall larkspur whose range spans much of the North American Cordillera from Alaska and British Columbia down through the Northern Rockies, with populations also present in the Cascades and Sierra Nevada. Surprisingly, this group also includes *D. exaltatum* Aiton, which occurs broadly in eastern North America and exhibits a disjunct pattern between the Appalachian and Ozark Mountains (Mohn *et al.*, 2021). The two populations of *D. glaucum* group monophyletically in the *caster* topologies, but *D. glaucum* from Alaska groups with the *D. brachycentrum* in the concatenation trees. Members of this clade vary in stature (the subarctic *D. brachycentrum* is short (20–50 cm) compared with *D. glaucum* and *D. exaltatum* (70–200 cm)) but are united by having glabrous to glaucous stems, moderate-length spurs (9–17 mm), paler flowers, and tending to occur in moist and rocky habitats, in contrast to morphologically similar, but dry adapted members of the CAL clade.

The ENA biogeographic region is currently represented by a single species, *D. carolinianum* Walter, whose range begins east of the Rocky Mountains. Inclusion of the remaining *Delphinium* taxa east of the Rockies is a vital next step that will help redefine a clade of species from eastern North America and complete the phylogeny of *Delphinium* sect. *Diedropetala*, likely clarifying the proper relationship between *D. exaltatum* and the other ALCAN species.

The SW clade is represented by eight taxa in our study which share a biogeographical distribution throughout the southwestern United States. This includes *D. scopulorum* A. Gray, *D. geraniifolium* Rydberg, *D. alpestre*, *D. robustum*, *D. ramosum*, *D. andesicola* Ewan, *D. sapellonis* Tidestrom, and *D. novomexicanum* Wooton. Notably, *D. ramosum* and *D. alpestre* are both polyphyletic. We recover three separate clades of *D. alpestre*, each appearing sister to a clade of *D. ramosum*, or of the very morphologically similar *D. robustum*. We investigate this ‘replicated divergence’ pattern in greater detail below. This group is united by all taxa having tall stature (50–200 cm), large leaves (5–15 cm wide), and large inflorescences (20–90 flowers), with the exception of *D. alpestre*, which forms a cushion habit (5–25 cm) and is only found above treeline.

Hybridization and introgression

Many *Delphinium* species overlap in both geography and phenology in western North America, where numerous putative hybrids have been described. Results of our f-branch analyses confirm that genomic admixture is widespread and common both within and between the major clades of *Delphinium* that we identified (Fig. 3; Table S2). Admixture is most common between taxa within the same geographically structured clades, or between clades with overlapping geographic ranges. These patterns are

consistent across alternative species tree hypotheses as well (Figs S5, S6). To avoid overinterpretation of these results, we focus below on describing the strongest signals of admixture, their taxonomic and geographic scope, and whether hybrids have been previously reported for the selected taxa based on observational or experimental evidence summarized in the FNA or Jepson Manual treatments of *Delphinium*.

Early diverged and unsampled lineages The strongest signal of admixture is observed in the early diverged lineages represented by several taxa co-distributed in California. The estimated admixture proportion between *D. patens* and *D. nudicaule* is *c.* 17%, while admixture between both of these taxa and *D. trolliifolium* is *c.* 6%. Given their positions in the tree, we cannot distinguish the directionality. These results are consistent across multiple replicate sampled individuals in both *D. nudicaule* and *D. trolliifolium*, lending increased confidence, although we have only one representative of *D. patens*. Genomic evidence of admixture between these species is consistent with reported observations that both species hybridize with *D. nudicaule* (Warnock, 1995). We do not find further evidence of *D. nudicaule* being admixed with other taxa in our dataset, despite reports of putative hybridization with *D. andersonii* and *D. nuttallianum*.

Although none of these early diverged lineages is strongly supported as a unique introgressive donor into other *Delphinium* taxa in our study, they do potentially provide evidence of other *unsampled* (or ghost) lineages that may be introgressive donors. This is most notable in the clade sister to *D. brachycentrum*, which includes the widespread taxa *D. glaucum* and *D. exaltatum*. This clade has the odd result of appearing admixed with every other sample in our phylogeny. Our best hypothesis to explain this involves introgression from a divergent unsampled lineage which we designate ‘unsampled lineage 1’ (Fig. 3). *Delphinium brachycentrum*, which lacks this signal of admixture, is the lone taxa in our study that also occurs in Asia. If it only recently dispersed to North America, that could explain why this introgression event only occurred into other North American members of this clade. An additional ancient introgression event may have occurred into the CAL clade, involving *D. cardinale* and/or *D. parryi*, from an unsampled lineage we designate ‘unsampled lineage 2’. Because this admixture signal is observed with respect to all taxa except *D. patens*, we propose that it may be derived from an unsampled or ghost member of a clade that diverged between *D. patens* and *D. nudicaule*, and which like them, exists or existed in California.

Admixture between geographic clades In addition to the early diverged lineages which overlap in California, there is also evidence of introgression between several other major biogeographic clades, particularly among members whose ranges overlap. This includes introgression between GB1 and PNW, involving *D. aff. nuttallianum* from N Utah and *D. andersonii* from S Idaho. These taxa are reported to hybridize (Hickman, 1993), and our analyses suggest that introgression is largely unidirectional into *D. andersonii*. The widespread taxon *D. glaucum* also appears admixed with several lineages, consistent with reports that it

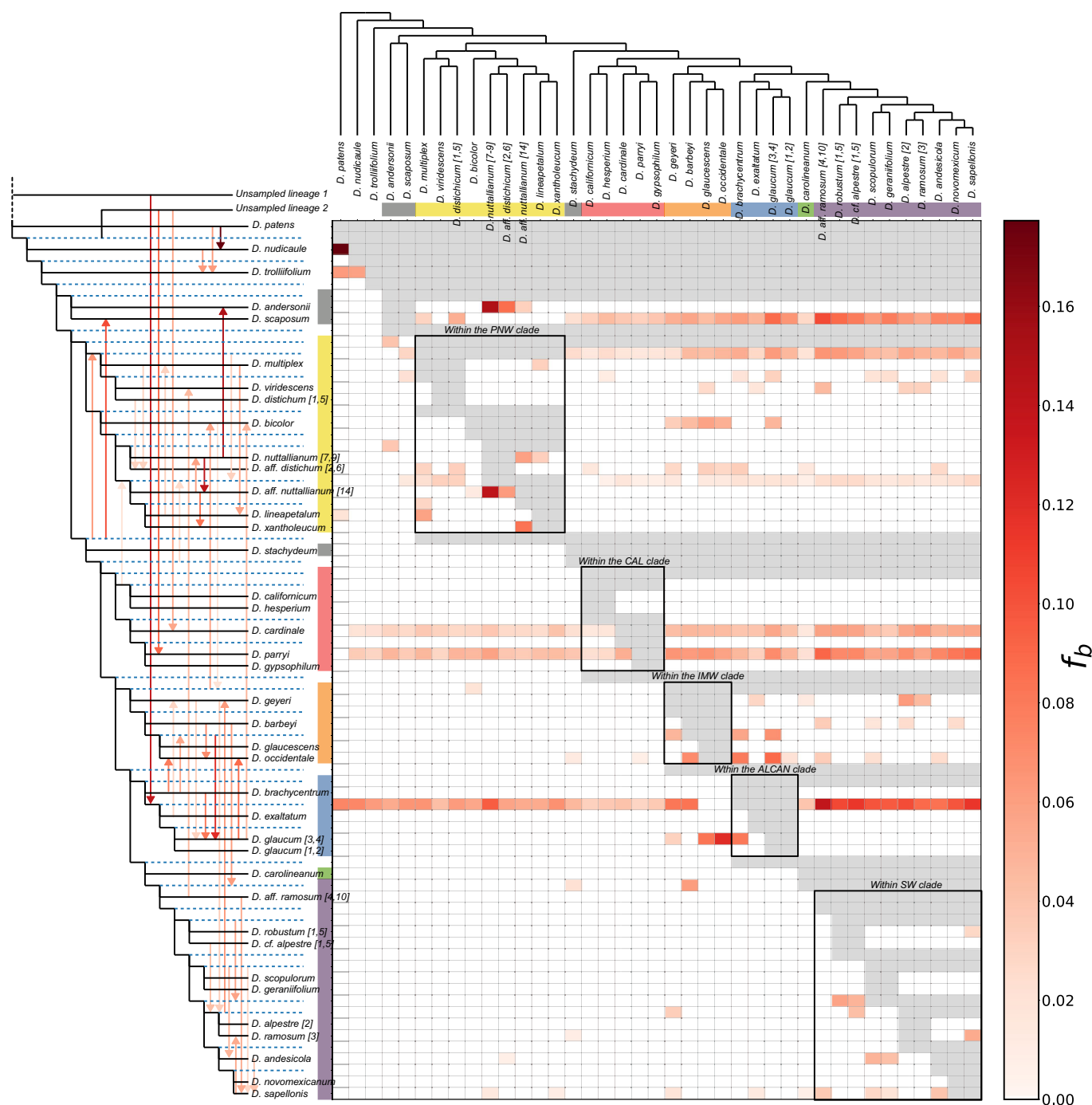


Fig. 3 Heatmap of f_b statistics summarizing admixture on the reference species tree of *Delphinium* sect. *Diedropetala*, with colored blocks indicating the major clades described above. Columns (x-axis) correspond to donor lineages, and rows (y-axis) to putative introgressive recipient lineages. Dotted lines on the y-axis indicate internal branches of the tree. Each cell shows the excess allele sharing between the donor and the descendants of one of the recipient lineages relative to its sister lineage. Broad bands indicate deeper introgression events, while patterns shared by one or few branches represent more recent and localized introgression. Arrows are drawn between branches on the left tree as a hypothesized summary of the strongest admixture signals, with arrow colors representing their mean f_b value. Gray boxes indicate tests that cannot be performed because they are inconsistent with the species tree topology used to generate these f_b statistics.

forms many types of hybrids (Warnock, 1995). Our genomic results confirm introgression with its close relative *D. brachycentrum*, as well as with *D. occidentale* from the IMW clade. The latter has been described as a hybrid between *D. glaucum* and the

IMW taxon *D. barbeyi*. Although *D. occidentale* appears admixed with both species, it also shares affinity with *D. glaucescens* and perhaps additional admixture with other taxa. Thus, our results suggest *D. occidentale* has a more complex origin that deserves

further investigation. Another taxon that appears admixed with *D. glaucum* is *D. bicolor*, a widespread taxon from the PNW clade, which has not been previously reported to hybridize. We do not observe admixture between *D. glaucum* and *D. stachydeum* from the GB2 clade, despite reports of hybrids between these taxa.

Further evidence of introgression between divergent clades is observed between members of the SW clade and members of both PNW and IMW, but none of these admixture proportions is particularly high. Although our f-branch results appear to support frequent admixture between divergent biogeographic clades, this pattern is largely driven by a few widespread species that hybridize abundantly, notably: *D. nuttallianum* (PNW clade); *D. barbeyi* (IMW clade); and *D. glaucum* (ALCAN clade). The ENA and CAL clades are relatively isolated from these widespread taxa, both geographically and ecologically, and are conspicuously absent of across-lineage introgression from these frequently hybridizing taxa.

Admixture within geographic clades There is also abundant evidence of admixture between species within each biogeographic clade. For example, within the PNW clade nearly every taxon is admixed with either *D. multiplex* or *D. aff. nuttallianum*; in the IMW clade, most taxa are admixed with *D. barbeyi*; and in the SW clade, there is a low level of admixture between most lineages. Given that these same taxa are widely involved in introgression between the biogeographic clades, this suggests that alleles from nearly any species are capable of being introgressed into another. An exception is the CAL clade, which shows little evidence of introgression between close relatives. This clade is notably characterized by high variation in flower color, suggesting that reproductive divergence may play an important role in reducing interspecies gene flow in this clade.

Discussion

Phylogeny and higher-level taxonomy in *Delphinium* sect. *Diedropetala*

We generated a large-scale genomic ddRAD-seq dataset to resolve phylogenetic relationships among populations and species spanning 34 taxa in the North American clade *Delphinium* sect. *Diedropetala*. Our phylogenetic results recovered most populations within taxa as monophyletic and grouped all taxa into eight consistently recovered higher-level clades that are strongly associated with biogeographic regions. We applied a range of phylogenetic methods and data filtering approaches to examine potential impacts of missing data or methodological biases. The primary disagreement among these results involves the order in which the eight major clades diverged across the backbone of the tree. These relationships received low support across all trees. This is consistent with the results of f-branch statistics, which indicate abundant hybrid introgression between species both within and between many of these clades. Our new phylogenetic hypotheses for the relationships within *Delphinium* sect. *Diedropetala* present a significant advance for understanding the history

of this North American plant clade by highlighting the dominant role of geography and hybridization in shaping its diversification.

Our phylogenetic results conflict with existing higher-level taxonomy in *Delphinium* sect. *Diedropetala*, including the current FNA treatment. Although our study does not include complete taxon sampling, we have included at least one representative of each subsection in the FNA treatment, allowing us to assess the concordance between molecular vs morphological similarities. Eight out of nine of the subsections in the FNA are recovered as polyphyletic in our phylogenies, representing discordance for every subsection for which we have more than one sample. This highlights the need for an updated taxonomy for this section in light of genomic evidence.

Taxonomic revision

We have assigned Phylocode-based taxon names for the six higher-level clades identified in this study that are represented by more than one species and generated morphological descriptions for each, highlighting morphological and ecological features that unite them (Notes S1; Fig. 2b). These formal clade definitions will aid future work aimed at re-examining our taxonomic hypotheses in light of new evidence. We designate Cordillerana n. cl. n. as the smallest clade containing *D. glaucum*, *D. exaltatum*, and *D. brachycentrum*, and which corresponds to the clade geographically centered in Alaska and Canada in our analyses. Columbianum n. cl. n. corresponds to our clade centered in the Pacific Northwest, and is defined as the smallest clade containing *D. lineapetalum*, *D. multiplex*, *D. xantholeucum*, *D. nuttallianum*, *D. distichum*, *D. bicolor*, and *D. viridescens*. Intermontana n. cl. n. is the smallest clade containing *D. barbeyi*, *D. glaucescens*, *D. occidentale*, and *D. geyeri*, and corresponds to our clade that is biogeographically centered in the Intermountain West. Sudmontana n. cl. n. corresponds to our Southern Rocky Mountains clade, and is the smallest clade containing *D. sapellonis*, *D. novomexicanum*, *D. andesicola*, *D. geraniifolium*, *D. scopulorum*, *D. robustum*, *D. ramosum*, and *D. alpestre*. Californica n. cl. n. corresponds to our clade centered in California and is defined as the smallest clade containing *D. cardinale*, *D. californicum*, *D. hesperium*, *D. gypsophilum*, and *D. parryi*. Finally, Basinensis n. cl. n. is defined as the smallest clade containing *D. scaposum* and *D. andersonii*, corresponding to our clade associated with the Great Basin.

Taxonomic clarification in light of admixture Our analyses also help to resolve long-standing disagreements concerning the taxonomy of multiple species. For example, we included multiple samples of *D. occidentale*, a species recognized by Ewan (1945) and Holmgren *et al.* (2012), but which Warnock (1995) describes as a hybrid between *D. glaucum* and *D. barbeyi*. Our phylogenetic results (Fig. 1) show *D. occidentale* as distinct from both *D. barbeyi* and *D. exaltatum*, and most closely related to *D. glaucescens* with 100% bootstrap support. In addition, our admixture results (Fig. 3) show that while *D. occidentale* is an introgressive recipient of both *D. glaucum* and *D. barbeyi*, it also appears to be an introgressive donor to multiple taxa within the

PNW clade. Given that nearly all taxa in our study show some evidence of admixture, we propose that *D. occidentale* is as deserving as any other taxon of species status. Our replicate population sampling, multiple phylogenomic analyses, and admixture analyses together provide multiple lines of genomic evidence to support previous analyses based on RAPD markers (Li *et al.*, 2002) and toxic alkaloids (Gardner *et al.*, 2002) that distinguish these three taxa from each other (for further discussion, see *D. occidentale* description in Holmgren *et al.*, 2012).

There is also disagreement across floras concerning the distinction between *D. ramosum*, *D. robustum*, and *D. alpestre* (Ewan, 1945; Weber & Wittmann, 2012; Ackerfield, 2022), a complex of species native to Colorado and New Mexico that are grouped into our SW clade. While *D. alpestre* only occurs above treeline and has a uniquely small stature, *D. ramosum* and *D. robustum* occur in subalpine communities and are larger and morphologically similar to each other, to the extent that *D. robustum* is sometimes not recognized as a distinct taxon, and instead lumped within *D. ramosum* (e.g. Ackerfield, 2022). Importantly, on alpine summits where populations of *D. alpestre* are found, there is almost always a corresponding population of one of these subalpine taxa at a lower elevation on the same mountain, leading some taxonomists to previously describe *D. alpestre* as a variety of *D. ramosum* (*Delphinium ramosum* Rydberg var. *alpestre* (Rydberg) W. A. Weber; Warnock, 1995).

Our dataset includes multiple sampled populations of each of these three taxa. In phylogenetic analyses using our full dataset, both *D. ramosum* and *D. alpestre* are polyphyletic, with the high elevation *D. alpestre* resolved into three different clades (Fig. 1). This pattern, along with samples from the putatively mixed Dalp-4 population being split across two clades, made it difficult to assign proper placement of *D. alpestre* in the species-level tree. Based on *D. alpestre*'s type locality, we refer to the Dalp-2 population as true *D. alpestre* and the other populations exhibiting the distinct high elevation diminutive phenotype as *D. cf. alpestre*. Similarly, we use the designations *D. ramosum* and *D. robustum* for the two populations exhibiting the lower elevation phenotype in the corresponding geographic region where their types were originally described. We refer to the final *D. ramosum*-like population, found in the southwestern San Juan Mountains, as *D. aff. ramosum*. Using a more densely subsampled sequence alignment (24 samples $\times$ 780 187 bp with 22.5% missing data) generated for this subset of taxa, while using the CAL clade as a closer outgroup, we inferred an additional ML concatenation tree which further confirmed the phylogenetic relationships observed in the larger tree (Fig. 4a). This phylogeny indicates that populations of *D. alpestre* are less related to each other than they are to populations of either *D. ramosum* or *D. robustum* at lower elevations nearby. As with the broader phylogenetic patterns in *Delphinium* sect. *Diedrope-tala*, relationships among populations within this clade match closely with geography, grouping into a northern, southeastern, and southwestern clade. These geographical clades align well with previous biochemical analysis that identified a northern and southeastern clade chemotype as distinct from a southwestern clade chemotype (Cook *et al.*, 2017).

We also examined patterns of ancestry among these populations using PCA performed on 2068 unlinked SNPs (0% missing data) from the 18 samples in this species complex. This also grouped samples into three distinct clusters corresponding to the same clades recovered in our phylogenetic results (Fig. 4c). However, when we add additional taxa representing other members of the SW clade, and putative admixed taxa from the IMW clade, we find evidence that their relationships are shaped by admixture. A final PCA was performed on 1792 unlinked SNPs (0% missing) across 39 samples from 8 taxa (Fig. 4d). Here, the southeastern and southwestern clades from this complex both appear intermediate between a cluster formed by all other members of the SW clade, and a cluster representing *D. geyeri* from the IMW clade. While this suggests admixture may contribute to the phylogenetic patterns in this group, it does not explain the replicated pattern of morphological divergence between *D. cf. alpestre* and its closest relative within each sub-clade, as they do not differ in their patterns of admixture. Further work is needed to investigate the role of introgression, plasticity, and adaptation in driving this replicated alpine divergence pattern.

Assessing an abundance of admixture

Although hybridization has been frequently observed and reported in North American *Delphinium*, our study is the first to use genomic evidence to validate many of these hypotheses by investigating signatures of admixture as an evolutionary consequence of hybrid introgression. Our results, based on patterns of allele sharing from thousands of genome-wide discordant SNPs examined in the context of phylogenetic hypotheses, show abundant evidence that hybrid introgression has occurred throughout the diversification of North American *Delphinium*. Many of the strongest signals of admixture that we detected are consistent with reports of hybridizing taxa listed in regional flora (e.g. *D. patens* $\times$ *D. nudicaule*; *D. nuttallianum* $\times$ *D. andersonii*; Hickman, 1993; Warnock, 1997); however, many other pairs of taxa that have been reported to hybridize do not show signals of admixture in our genomic results (e.g. *D. nudicaule* $\times$ *D. nuttallianum*). This may indicate that some hybrids are infertile, or it may reflect that introgression is sometimes geographically localized, and occurred outside of the range of populations sampled in this study. Finally, we also detected evidence of admixture between many taxa that have not been previously reported to hybridize (e.g. *D. glaucum* $\times$ *D. bicolor*), identifying case studies for future investigations.

Not every hypothesized introgression event reconstructed from our f-branch results in Fig. 3 is likely to be true. Both analytical and bioinformatic errors can inflate these estimates, and extinct and unsampled lineages may also contribute to these patterns. We suspect that some of the hypothetical introgression events may represent false positives that can arise from nonindependent signals of admixture among multiple hybridizing taxa (Eaton *et al.*, 2015; Malinsky *et al.*, 2021). Because each test examines only one quartet of taxa at a time, and different sets of taxa may

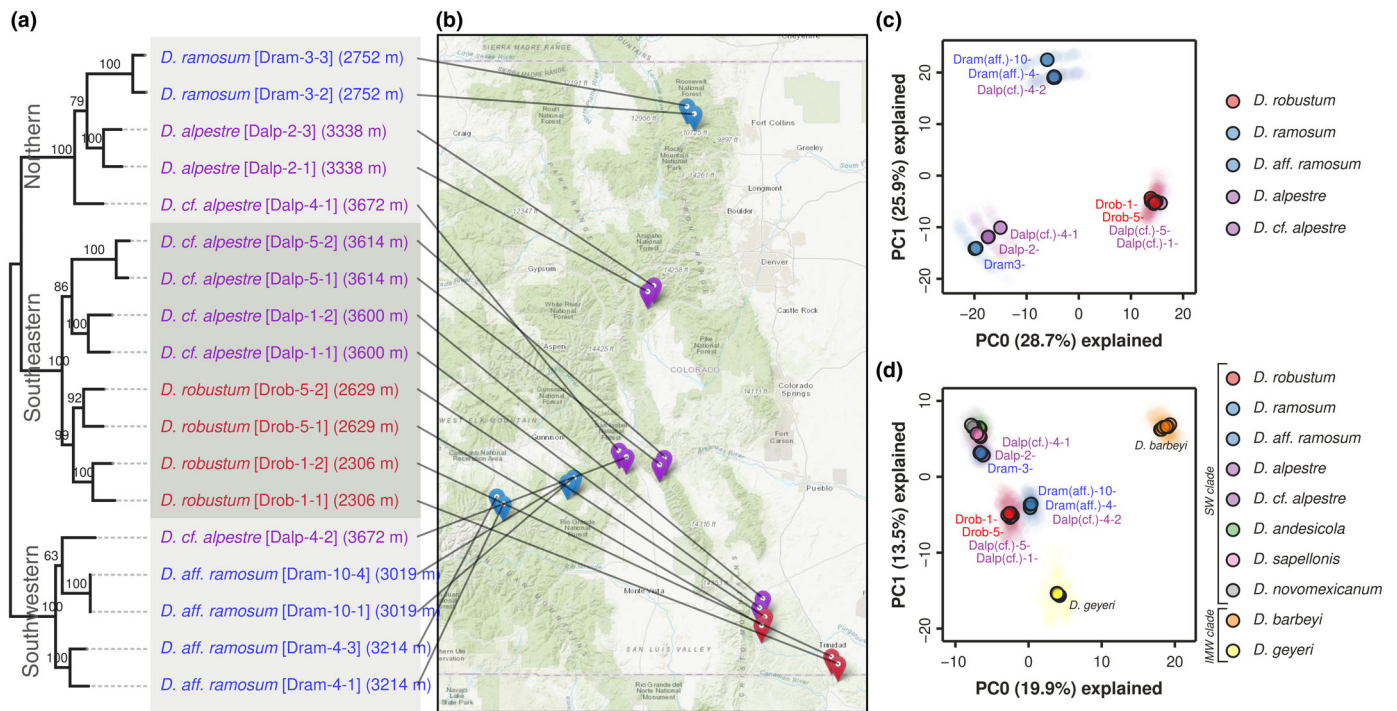


Fig. 4 Hybrid complex of *Delphinium robustum*, *Delphinium ramosum*, and *Delphinium alpestre* in the Southern Rocky Mountains. (a, b) Three distinct clades are strongly supported in phylogenetic results, grouping samples by geography rather than by species determinations based on morphology. (c) A genomic principal component analysis (PCA) similarly groups samples by geography. Low-opacity points show variation over replicate runs using randomly sampled subsets of unlinked single-nucleotide polymorphisms (SNPs). Each sample is represented by a centroid point. Labels show individual sample names, or population names when points overlap. (d) A genomic PCA with the inclusion of other members of the SW clade, and potential introgressive donors from the more distantly related IMW clade (*Delphinium geyeri* and *Delphinium barbeyi*). IMW, Intermountain West; SW, Southwestern United States.

share different sets of loci in RAD-seq datasets, even methods like the f-branch statistic that aim to account for the nonindependence among tests may yield false positives. Nevertheless, ABBA-BABA tests have been widely applied to RAD-seq datasets (Consortium *et al.*, 2012; Eaton & Ree, 2013; Eaton *et al.*, 2015; McCluskey & Postlethwait, 2015; Vargas *et al.*, 2017) and do not typically lead to estimates of introgression nearly as abundant as observed in this study, suggesting that our results are not an artifact of using anonymous genomic markers alone. Similarly, while bioinformatic errors related to paralogy estimation and variant calling can also inflate estimates of introgression, we have no reason to expect these are inflated in our dataset. For example, *Delphinium* is not characterized by frequent polyploidy. Karyotypes have been well studied in *Delphinium* (Lewis *et al.*, 1951; Legro, 1961; Turner, 1992) where most species are diploid with a haploid chromosome number of eight ($2n = 16$). Among the 27/34 taxa in our study for which data are available, only *D. exaltatum* is polyploid ($4n = 32$), while all other taxa have been reported as diploids ($2n = 16$), although four have been reported to sometimes also occur in tetraploid populations, or as tetraploid cultivars (*D. gypsophilum*, *D. carolinianum*, *D. glaucum*, and *D. geyeri*). With the exception of *D. glaucum*, none of these occasionally polyploid taxa is among those most widely implicated in introgression in our results.

We have primarily interpreted our admixture results in the context of introgression. However, another potential consequence of hybridization is hybrid speciation. This process would be better modeled through alternative methods, such as network analyses, that do not rely on a species tree hypothesis. In this study, we are limited by the information available in short *de novo* assembled RAD-seq loci, which individually do not generally contain sufficient information to infer reliable gene trees to use as input to common phylogenetic network inference tools (Solís-Lemus & Ané, 2016). This was the case for our dataset, where we recovered many SNPs in total, but very few SNPs per individual locus. We were also unsuccessful in assembling RAD-seq loci mapped to the chloroplast, which could otherwise provide a point of comparison for investigating patterns of introgressed ancestry. Although our admixture analyses have revealed many new insights into the history of hybridization in *Delphinium*, we expect much more remains to be discovered in future studies. These would benefit most by increasing taxon sampling; generating and using one or more nuclear reference genomes for whole-genome resequencing to improve paralogy estimation and comparison with plastome phylogenies; and employing more advanced methods for testing introgression hypotheses that can model phylogenetic networks and make use of spatially structured patterns along chromosomes.

A Western North American *Delphinium* Syngameon

The term syngameon describes a set of species that are connected by hybridization (Boecklen, 2017). Well-known examples, such as oak trees, have long been studied as both an intractable taxonomic challenge and fascinating evolutionary phenomenon, and recent genomic studies have only confirmed the complexity of this system (Eaton *et al.*, 2015; Hipp *et al.*, 2020; Ribicoff *et al.*, 2025). A notable feature of syngameons is that not all species hybridize evenly, yet through a network of connections – with a few species functioning as super-hybridizing nodes – it is possible for genes to be exchanged, at least in theory, between any two species in the complex (Boecklen, 2017). This is precisely the pattern that we observe in North American *Delphinium*. Hybridization and admixture is common among co-distributed species, yet many taxa have become differentiated across geographic and ecological space, likely due to a combination of selection and drift acting on unique and shared alleles. In addition, several widespread species, such as *D. nuttallianum*, *D. glaucum*, and *D. barbeyi*, hybridize with species from multiple disparate geographic regions, potentially facilitating the exchange of alleles over greater geographic distances. While this presents a challenge to many traditional methods in evolutionary systematics, such as reconstructing the biogeographic history of species, it presents many exciting opportunities as well, for reconstructing the biogeographic history of both neutral and functional alleles beyond the constraints of species barriers.

Conclusions

Our phylogenomic analyses present new hypotheses and insights into the evolutionary history of this rapidly radiated clade and demonstrate that the divergence history of this group is more complex than morphology alone reveals. We discover an overwhelming signal of geographically structured ancestry, with hybrid introgression connecting species both within and between distinct geographic regions. Future work should include the remaining unsampled taxa to better address questions concerning the origin and early dispersal history of *Delphinium* upon its arrival in North America and as it dispersed east, including diversification patterns linked to climatic, topographical, and pollination shifts. Toward this goal, our phylogenetic revision of *Delphinium* sect. *Deidropetala* represents a major advance for understanding the evolutionary history of this enigmatic North American plant clade.

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Competing interests

None declared.

Author contributions

JBM, DARE and DC planned and designed the research. JBM and DC conducted fieldwork. BSB and KT conducted molecular lab work. JBM and DARE analyzed the data and wrote the manuscript.

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Data availability

Sequence data are available on the NCBI Sequence Read Archive under project identifier PRJNA1416459. Accession numbers for samples used in this study can be found in Table S1. Phylip formatted sequence alignment files and Newick formatted tree files are available on DRYAD: [10.5061/dryad.8cz8w9h5p](https://doi.org/10.5061/dryad.8cz8w9h5p).

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 A rooted phylogenetic tree for 150 samples of *Delphinium* sect. *Diedropetala* inferred from the ‘min4’ SNP dataset using *caster*.

Fig. S2 A rooted phylogenetic tree for 150 samples of *Delphinium* sect. *Diedropetala* inferred from the ‘min4’ sequence alignment using *raxml-ng*.

Fig. S3 A rooted phylogenetic tree for 150 samples of *Delphinium* sect. *Diedropetala* inferred from the ‘min10’ sequence alignment using *raxml-ng*.

Fig. S4 ASTRAL species tree inferred from 3121 gene trees from individual RAD-seq loci.

Fig. S5 f-branch analysis of admixture performed on the *raxml-ng* ‘min4’ species tree hypothesis.

Fig. S6 f-branch analysis of admixture performed on the *raxml-ng* ‘min10’ species tree hypothesis.

Notes S1 New clade definitions and supplemental methods.

Table S1 *Delphinium* sect. *Diedropetala* plant accessions used in this study with GPS coordinates and sample identifiers for vouchers stored at the USDA-ARS Poisonous Plant Research Laboratory in Logan, UT, USA.

Table S2 *Dsuite* f-branch analysis results performed on the *caster* ‘min10’ 41-tip species tree hypothesis of *Delphinium* sect. *Diedropetala*.

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